Computational Modelling of Bioelectric Pattern Formation in Multicellular Systems

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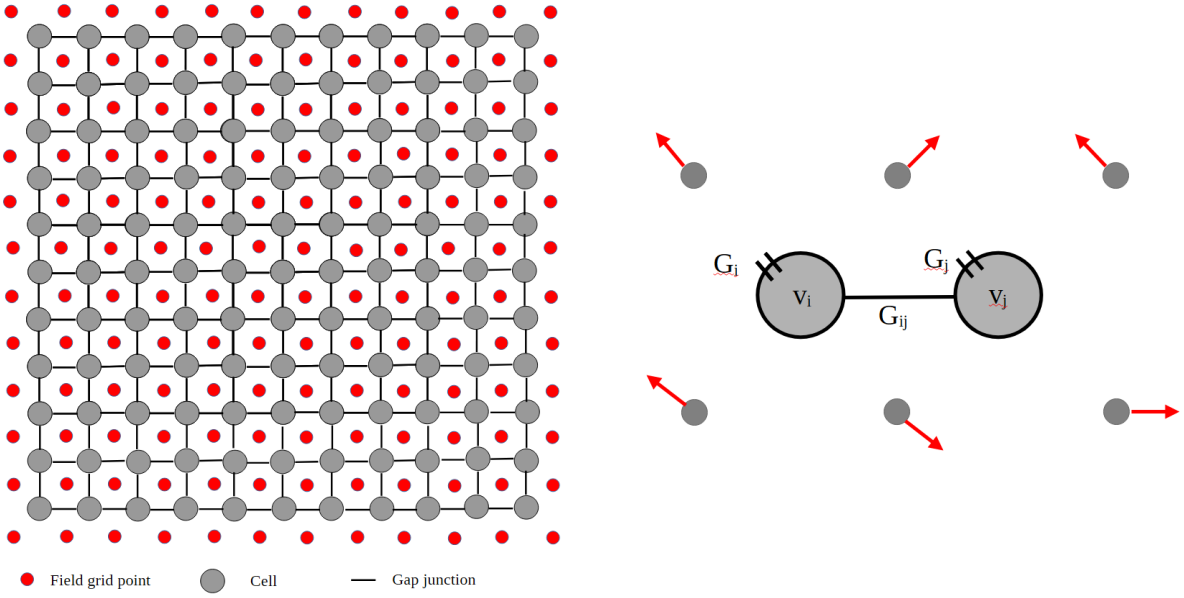
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Introduction

Great advances have been made over the last decades in genomics and molecular biology, enabling gene editing and a very precise understanding of a vast array of areas from evolution to genetics. However, despite these advances, genomics has not yet been able to explain large scale information in biological systems, [1],[2] such as the coordination of embryo development and body plan organisation. This has led to recent research demonstrating that biophysical properties of cells play a key role in high level coordination in biological systems. All cells, not just neural cells have actively maintained electrical properties, the most significant of which is the membrane potential, V_{mem}.[3] Cell's V_{mem} are actively maintained by a variety of ion channels, which allow diffusion of, or actively pump electrically charged ions across the cell membrane, to maintain a specific membrane potential, which differs by tissue and cell type. Studies have shown that this membrane potential acts on two levels: at the cellular level the membrane potential is a master biophysical regulator of cell behavior and phenotype, while on the tissue level the spatiotemporal patterns of of cell V_{mem} states form an instructional code that determines global system properties such as tissue type, limb formation and homeostasis.[4] This can even be visually seen with voltage imaging of developing frog embryos,[5] for example. Despite these advances, analysis of the information stored in the cell bioelectric networks, coordinated by V_{mem}, the cell coupling gap junctions, and the induced electrostatic fields, is in a much earlier stage. And a complete theory of how this information is created, maintained, and employed by cells to coordinate large scale behavior has not yet been formed. It was Burr[6] who, observing physics progressing from particle based formulations to field theories, first proposed a bioelectric field theory for biology, and the idea of voltage gradients guiding patterning in biology. However, it has taken many decades before this hypothesis has been tested and developed to a reasonable degree.
Many gene expression [7] and epigenetic[8] targets are known to lie downstream of changes in the bioelectric state of cells. In recent studies *in vivo* experiments have used spatially specific, artificially induced manipulation of membrane potentials to induce eye formation,[9] repair birth defects,[10] induce and reverse oncogene induced carcinogenisis,[11] and regrow normally non-regenerating frog legs.[12] Alongside these highly important specific discoveries, it is of great importance to have the ability to fully understand how cell membrane potentials and electric fields coordinate pre-patterning, and allow information storage and use on the multicellular scale. Further knowledge in this field could lead to improvements in fundamental understanding of biological organisms, and future medical treatments in a wide range of fields from regeneration to oncology.

To this end, I adapted the model from the recent computational paper **Field-mediated bioelectric basis of morphogenetic prepatterning** [3]. The model developed consists of a two dimensional lattice of biological cells. Each cell is equipped with a generalised polarising and depolarising ion channel, gap junctions connecting each cell to its four neighbors in the lattice, and an embedded electrostatic field, which allows a feedback mechanism between membrane potential and field excitation. Extracellular fluid and other tissue properties, which would also play a role in the real world, are ignored in this model. Here we analyse the model to understand it's information storing ability, and its characteristic behavior. The model has a field transduction mechanism where the ion channels (here the polarising channel, G_i and G_j in **Figure 1 (b)**) are field gated as well as voltage gated. This is implemented through the electric field vector (also shown in **Figure 1 (b)**) magnitudes from the four field grid points that surround a cell being averaged, to give the sensed field by the cell. This is based on observed electroreception phenomena in nature.[13], [14] Cells then radiate their electric field to the surrounding field grid points (red dots in **Figure 1 (a)**), with a distance set by the parameter termed field action range, R. The equilibrium potentials of the polarising and depolarising channels, E_p and E_d respectively, also play an important role in the model dynamics, which can be seen in the convergence behavior of the governing equations. The model is characterized by four key

parameters, that we seep over in our analysis: field waiting W, field transduction threshold B, field action range R, and gap junction strength G.



(a) 11 × 11 cell lattice **(b)** Two–cell model

Figure 1: Bioelectric cell lattice (a) electric field grid points (red dots), cells (grey circles), and gap junctions (black lines). (b) zoomed in on two cells, illustrating electric field vectors, gap junction, and polarisng and depolarsing ion channels.

Model equations

Here we lay out and explain the governing equations of the computational model.
Conductance of gap junction connecting cells i and j:

$$G_{ij} = \frac{2GG_r}{1 + \cosh\left(\frac{v_i - v_j}{v_0}\right)} \tag{1}$$

The dominant behavior comes from the criticality of the cosh function: 1/cosh(x). This results in the conductivity being highest when the membrane potential of cells i and j are closest, and weakest when the membrane potential of cells i and j are furthest apart. This is because when i and j are similar, $\frac{1}{\cosh(0)} = 1$, and hence G_{ij} is a maximum. And when i and j are far apart, G_{ij} is minimized. Therefore, cells with initially similar membrane potentials will synchronise exponentially, while cells with strongly different V_{mem} can remain seperate they will eventually equilibrate but this happens very slowly.

G_{p_i} conductance of polarising ion channel of cell i:

$$G_r \frac{dG_{p_i}}{dt} = -G_{p_i} + 2W\left(\sigma\left(-\sum_{j \in N_{F_i}} \frac{\|\vec{e}_j\|}{|N_{F_i}|} + B\right)\right) - 1 \tag{2}$$

G_r is a constant, reference value for scaling ion channel gap junction conductance. W is the field transduction weight. σ is the sigmoid transformation. $\boldsymbol{e}_j$ is the electric field vector at field grid point j. N_{fi} is the number of neighbouring grid points of cell i. This is a parameter which can be adjusted. B is the field transduction threshold. When the average field from the vectors, after being transformed by the weighted (W) sigmoid function, exceeds B, the G_p of the contained cell decreases, resulting in its gradual depolarization. Below the threshold, G_p increases, resulting in hyperpolarization. Thus, when

$B = 0$, all the cells depolarize, and when $B = \infty$ the entire tissue hyperpolarizes. For intermediate values of B, heterogeneous mixtures of states can appear.

Electric field vector at field grid point i:

$$\vec{e_i} = \frac{k_e}{\varepsilon_r} \sum_j \frac{q_j}{\|\vec{r}_{ji}\|^2} \delta(\vec{r}_{ji} - \sqrt{2}rR) \tag{3}$$

Equation for V_{mem} of cell i:

$$C\frac{dv_i}{dt} = -\frac{G_{p_i}(v_i - E_p)}{1 + e^{\frac{z(v_i - v_{th})}{v_T}}} - \frac{G_{d_i}(v_i - E_d)}{1 + e^{\frac{-z(v_i - v_{th})}{v_T}}} + \sum_{j \in N_{C_i}} (v_j - v_i)G_{ij} \tag{4}$$

Let

$$s(v_i) = \frac{1}{1 + e^{(\frac{-z(v_i - v_{th})}{v_T})}} \tag{5}$$

Where v_{th} is threshold voltage, and v_t controls the sharpness of the transition of the sigmoid function. With some algebra we can then write:

$$C\frac{dv_i}{dt} = -G_{p_i}s(v_i)(v_i - E_p) - G_{d_i}[(1 - s(v_i)](v_i - E_d) + \sum_{j \in N_{C_i}} (v_j - v_i)G_{ij}$$

Now if we ignore neighboring cells then $G_{ij} = 0$ (and we can drop the subscripts), and divide through by C. We can look at the steady state where $\frac{dv_i}{dt} = 0$ and we get:

$$0 = G_p s(v)(v - E_p) + G_d[(1 - s(v)](v - E_d)$$

Rearranging gives:

$$v = \frac{G_p s(v)E_p + G_d[(1 - s(v)]E_d}{G_p s(v) + G_d[(1 - s(v)]}$$

Now we can see that when $v << v_{th}$ we have $\frac{1}{1+e^{(large)}}$ so $s(v) \approx 0$. Therefore the membrane equation becomes $C\frac{dv_i}{dt} = -G_d(v - E_d)$ This has solution of the form:

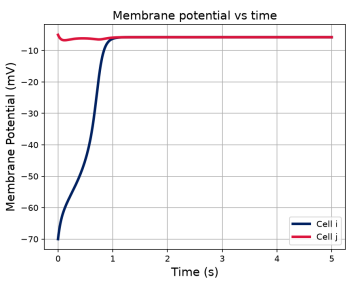
$$v(t) = E_d + [v(0) - E_d]e^{\frac{-G_d t}{C}} \tag{6}$$

So as t progresess the voltage is driven towards $E_d = -5mV$ (commonly used voltage), with characteristic time $\tau = \frac{C}{G_d} \approx 67ms$. When $v >> v_{th}$ the same reasoning and working leads to the membrane voltage being driven towards $E_p = -55mV$ with a characteristic time $\tau = \frac{C}{G_p} \approx 100ms$.

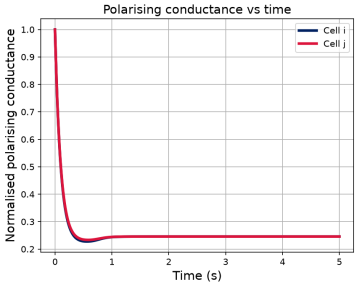
Two–cell model simulations

A computational simulation was developed to implement the above model. The model geometry was defined with two cells, connected by one gap junction, and surrounded by 6 electrostatic field grid points **Figure 1 (b)**. The timestep for the simulation was set to $dt = 1ms$, and the simulations ran for $\Delta t = 5s$. The above equations were coded as separately, with independent functions for the electric fields produced by each cell, the net electric field from both cells at each grid point, and to calculate the magnitude of the electric field sensed by each cell. separate equations were also written for the opening probabilities of the polarsing and depolarising ion channels. Although the conductance of the depolarisng channel remains constant, its opening probability is still dependent on the cell's membrane voltage. These functions correspond to the exponential components in the denominator of **equation (4)**. The SciPi function $expit(x)$ was used here instead of exponential numpy functions, as a more numerically stable approach. A coupled ODE solver was then implemented using SciPi, which was fed the required initial conditions.

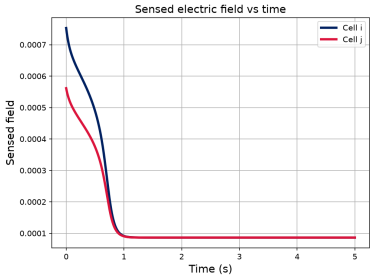
To begin with, we plotted the membrane potentials (V_{mem}) for both cells, along with the behavior of the polarizing conductance channels, and the electric fields sensed by each cell. Simulations were performed with the two cells initiated at different V_{mem}, one polarised/hyperpolarised and the other depolarised. The five parameters which controled the behaviour of this simplified model are the equilibrium potentials of polarising and depolarising ion channels (E_p and E_d respectively), threshold voltage v_{th}, field transduction weight W and field transduction threshold B. The combination of these paramters, and the relatively large range of values they can take gives a very large number of inital combinations. Despite this the model tended towards two different types of behaviour: either the membrane potentials of the two cells reached an equilibrium, along with the sensed fields from each cell **Figure 2 (a) – (c)**, or the membrane potentials and the sensed fields did not converge **(d) – (f)** of **Figure 2**. The latter occurred when the difference between E_p and E_d was too big, or for specific ranges of B for a given set of the other parameters. For example, from $B = 0$ to $B = 0.01$ both cells depolarise, while setting $B = 0.03$, the cells hyperpolarise. At $B = 0.02$ the cells remain roughly at their initial states and do not converge towards one value (for values of $W = 100$, $E_p = -55$, $E_d = -5$, $v_i = -55.0$, $v_j = -5.0$. In the first scenario when the membrane potentials and sensed fields converge, the convergence was largely due to the values of E_p and E_d, which is to be expected looking at equation **(6)** and associated reasoning. In both cases illustrated in **Figure 2**, the conductance of the polarising channels remained similar, although with slightly more interesting behaviour for the converging case.



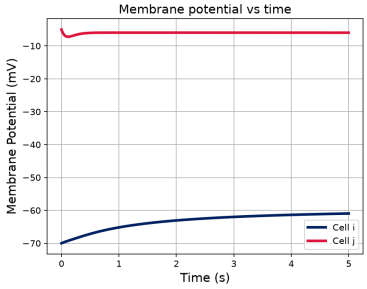
(a) Membrane potentials.



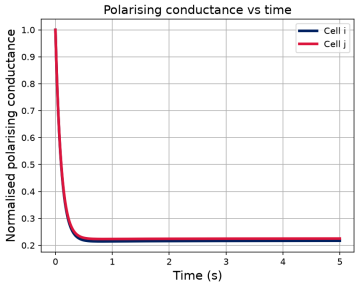
(b) Polarising channel conductance.



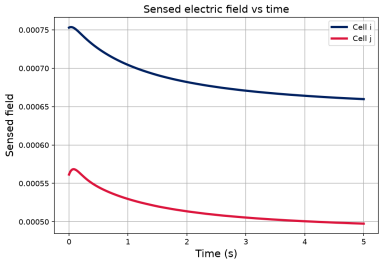
(c) Sensed electric fields.



(d) Membrane potentials.



(e) Polarising channel conductance.



(f) Sensed electric fields.

Figure 2: 3 main graphs showing characteristic behaviour of system's governing differential equations, for 2 cell model. (a) – (c) with $E_p = -55$, $E_d = -5$, $z = 3.0$, $v_{th} = -27$ and $G = 0.01$. For (d) – (f) $E_p = -70$, $E_d = -5$, $z = 3.0$, $v_{th} = -27$ and $G = 0.01$, here graphs have separated and do not converge because E_p, E_d, $G = 0.03$ and $z = 3$ determine convergence behaviour. Complexity plot for (a) – (c) and (d) – (f), are Figure 3 (a) and (c) respectively.

Complexity analysis: Two–cell model

Before expanding to a larger lattice of cells, analysis was performed on the 2 Cell model, to analyse simple behaviour and confirm analysis methods. To quantify the information in the system, Tononi-Sporns-Edelman (TSE) complexity analysis was used. TSE was developed in the context of neural systems, but here we apply it to a lattice of non-neural cells. It is a measure of how much a

system contains parts that behave independently, while still being strongly integrated into a coherent whole. Here it is quantifying the compromise between integration and segregation of the V_{mem} states and (when using larger lattice of cells) patterns. This means that when the model is completely integrated with the cells behaving synchronously, or when the cell's behavior is completely segregated and not effecting one another, the TSE complexity is zero. TSE is maximized when there is a balance between integration and segregation, so that the cells behave independently but still effect each other. We used TSE defined as:

$$\mathrm{TSE}(X) = \sum_{j=1}^{n/2} \left[\langle H(X_{k_j}) \rangle - \frac{k}{n} H(X) \right] \tag{7}$$

where

$$H(X) = -\sum_{x \in X} p(x) \log p(x) \tag{8}$$

is the Shannon entropy. Here, $H(X)$ is the joint entropy of the entire system of cells, while $\langle H(X_k^j) \rangle$ is the average joint entropy of all subsets containing k cells. Here for the 2 cell model, this simplifies to

$$TSE = \frac{1}{2}(H(X_1) + H(X_2) - H(X_1, X_2))$$

which is plotted as a function of W and B in **Figure 3**. Later when the number of cells is increased, we also plot TSE as a function of the field range R, but this can't be done with the 2 cell simplified case.

Each cell has a membrane voltage which changes over time, V_{mem}. For the TSE calculation these membrane voltages were converted into binary states: 0 or 1, for the cell's V_{mem} being below or above its median voltage respectively. Therefore, in this implementation each cell's threshold is determined by its own voltage trajectory rather than a fixed threshold. This method was changed in later implementation when the model is expanded to a lattice, and a fixed physiologically relevant threshold is set ($v_{th} = -27mV$). An entropy calculation function counts the number of times each cell was in each state, creates the probabilities of each, and then calculates the Shanon entropy. The main TSE function then calls this function, and along with the binarized cell states, computes the TSE complexity. It should be noted that for this simplified 2 cell model, TSE is equivalent to half the mutual information of the system.

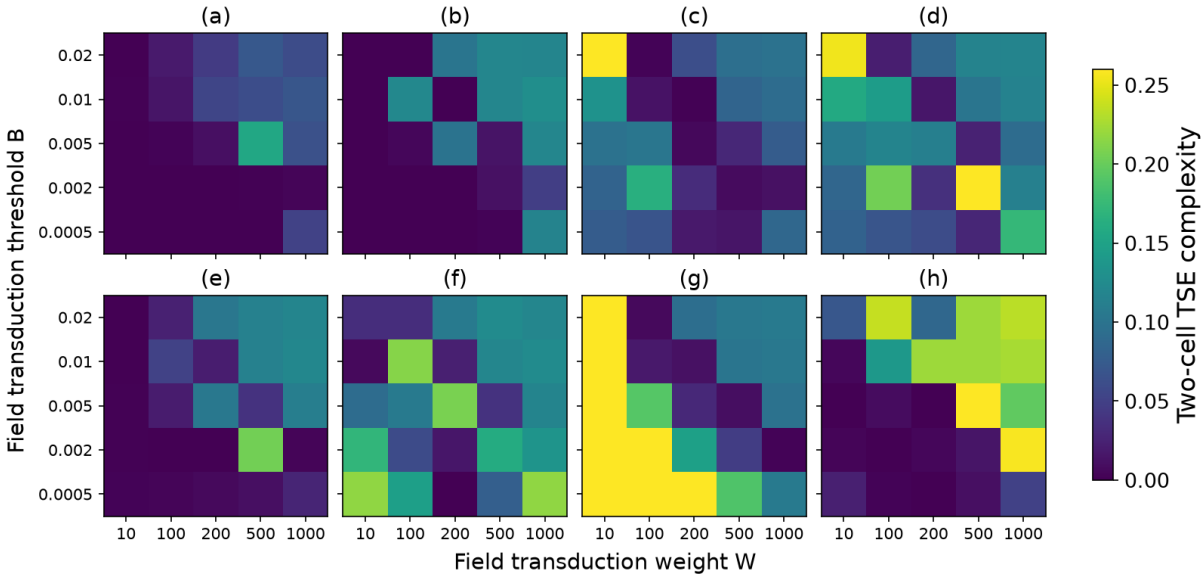


Figure 3: 2 cell TSE complexity heatmap plots, for changing values of W and B. Each plot has different E_p, E_d, v_{th}, z and G values. **(a)** $E_p = -55$, $E_d = -5$, $v_{th} = -27$, $z = 3.0$, $G = 0.1$. **(b)** $E_p = -55$, $E_d = -15$, $v_{th} = -27$, $z = 3.0$, $G = 0.1$. **(c)** $E_p = -70$, $E_d = -5$, $v_{th} = -27$, $z = 3.0$, $G = 0.1$. **(d)** $E_p = -70$, $E_d = -15$, $v_{th} = -27$, $z = 3.0$, $G = 0.1$. **(d)** $E_p = -70$, $E_d = -15$, $v_{th} = -20$, $z = 3.0$, $G = 0.1$. **(e)** $E_p = -70$, $E_d = -15$, $v_{th} = -27$, $z = 3.0$, $G = 0.8$. **(f)** $E_p = -70$, $E_d = -15$, $v_{th} = -20$, $z = 3.0$, $G = 0.1$. **(g)** $E_p = -55$, $E_d = -5$, $v_{th} = -27$, $z = 5.0$, $G = 0.1$. **(h)** $E_p = -55$, $E_d = -5$, $v_{th} = -20$, $z = 1.0$, $G = 0.1$.

From **Figure 3 (a) – (c)**, it can be seen that decreasing E_d, and therefore decreasing the difference between E_d and E_p, has little effect on the TSE plots, while decreasing E_p increased complexity for lower values of W, and especially at higher values of B. In **(d)** decreasing E_p and E_d increased the complexity for lower values of B. Significantly increasing G from **(d)** to **(e)** concentrated the complexity in high values of W and B, while reducing it elsewhere. Decreasing v_{th} in **(f)** in relation to **(a)** increased complexity uniformly across W and B sweep, and visa versa. Higher values of z in **(g)** increased complexity for low W and high B values. In **(h)** increasing v_{th} from -27 to -20 mV, significantly decreased complexity except at specific regions for high values of W and lower values of B, creating an unusual pattern, relative to **(a)**.

The above behavior follows the expected biophysics of the model. It can be seen from the analytical analysis above that E_p and E_d are the preffered polarised/depolarised membrane states, so changes in these alteres how easily cells can transition between states, thus altering how heterogeneous the cell behaviour can be. Higher W increases the strength of the field feedback, so changing parameters which effect membrane potential or field sensing will generally have a bigger effect when W is large. B shifts the field-transduction sigmoid relative to the sensed field, so changing E_p and E_d which changes the voltages and therefore the fields can move the system into or out of a sensitive part of the sigmoid, and this likely explains why some effects appear at specific B values.

Increasing gap-junction coupling G pushes neighboring cells toward similar voltages, so stronger coupling therefore tends to synchronize the cells and suppress independent behavior, this can reduce complexity over much of the W and B parameter space. Strong field feedback which corresponds to large W, and an appropriate B, is then needed to overcome that homogenizing effect, which explains why complexity becomes concentrated in a smaller high W and B region. This makes sense and seems to be relatively independent of other parameters. It also aligns with recent experimental findings[15] demonstrating that gap junction coupling between cells suppresses electrical fluctuations by passively stabilising V_{mem}, and thus reducing complexity. And that disruption of this through oncogenic ion channels helps drive electrical volatility in cancer.

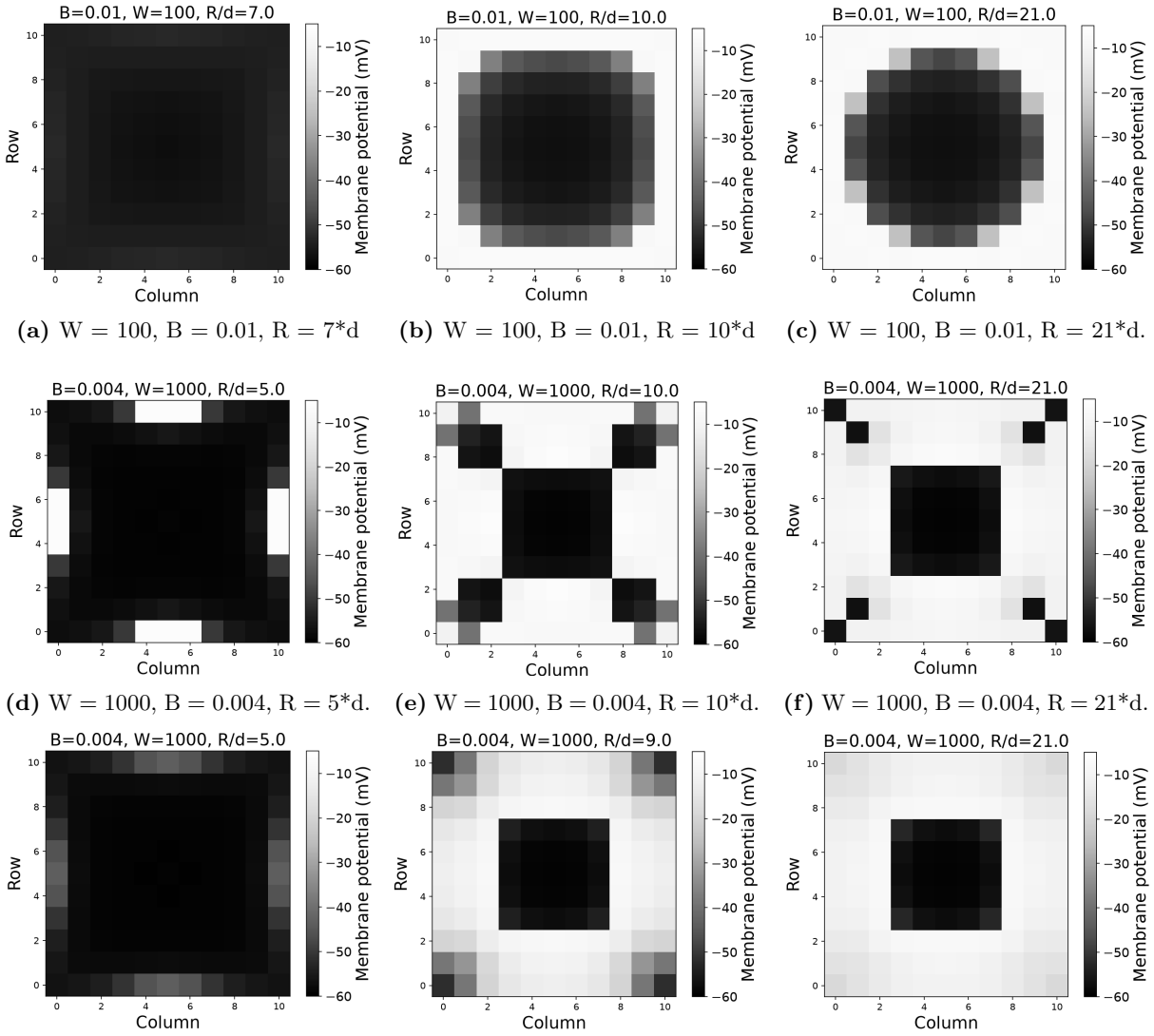
Larger z makes the voltage dependence steeper. So cells behave more like switches around V_{th}, strengthening the distinction between polarized and depolarized states. This can support more pronounced spatial patterns and hence higher complexity. However strong field weighting W and high threshold B likely combine to reduce this effect, which explians **Figure 3 (g)**. The Small z makes the response smoother and weakens this bistability, so it is understandable that complexity only survives in

a narrow region of W, B parameter space as seen in **Figure 3 (h)**. Since z shifts both sigmoids, in the dv_i/dt function along the voltage axis, this changes the balance between the two polarisng and depolarizing currents, specifically increasing V_{th} makes the polarizing channel relatively more influential and the depolarizing channel less influential. The reason the effect is at lower W is that since W controls the strength at which the sensed fields change the polarizing conductance, for high values of W the field feedback dominates and can override changes to intrinsic voltage gating dynamics. However, at lower W the field feedback is weaker and this results in the cells being more sensitive to voltage gating dynamics.

Full cell lattice model simulations

The model was expanded to an 11×11 cell lattice. This allowed a more fruitful analysis of the cell membrane potentials in a lattice of cells and facilitates investigation of spatiotemporal pattern formation, complexity measures, and dynamical nonlinear behavior of cells.
Expanding the model to an 11×11 cell lattice, most of the underlying biophysical equations remain the same, however the expanded geometry required differences in coding implementation and much more efficient code. There are two main biophysical differences that come with expanding the model, the first is that each cell is now connected to its 4 neighbors via gap junctions, except for edge cells which are connected to 3, and corner cells which are only connected to 2. The second is that we now have a new parameter, termed field action range R, this is the distance over which a cell can radiate its electric field (from its V_{mem}) to field grid points.
Whereas in the 2 cell model we defined each cell separately, here we used nested loops to automatically generate the coordinates of all 121 cells. To significantly save computational time, the model was adapted to precompute the geometry, creating two arrays containing the vector from each cell to each field grid point, and the distance to each field point. These arrays are then used throughout the simulation and this avoids the ODE solver recalculating the geometry thousands of times. For each field point in the 2 cell model python explicitly calculates the contribution from both cells. In the full lattice model, this was optimised using NumPy vectorisation to calculate the contributions from all 121 cells at all 144 grid points in a single expression. This is achieved through broadcasting, where $None$ introduces additional dimensions to the charge and distance arrays, which allows NumPy to automatically match each cell's charge and distance with its corresponding displacement vector. This eliminates the need for nested python loops making it computationally faster. Vectorisation was also used in summing the electric fields. To model the gap junctions, rather than hard coding every connection, a list of *neighbors* was constructed and pre-calculated, this list was looped through in the ODE solver. Finally, array slicing was used to split the state vector into two arrays containing the cell voltages and normalised polarising ion-channel conductances. An appropriately adapted *model* function was used to implement SciPy's *solve.ivp* to integrate the entire network.

Here we used RK45 integration to solve the system numerically. RK45 treats the model as continuous time ODEs, it chooses and varies its own internal time steps based on the rate of change of the solution, specifically the difference between the 4th and 5th order Rung-Kutta result, and uses this to construct a highly accurate Rung-Kutta estimate. This means that it constructs a continuous solution to the entire model, and simply produces the solutions at the requested time steps for the simulations (**Figure 4**). This differs from the original paper where they used explicit discrete updates with a pre-set time steps. For simple ODEs both methods should converge on the same solution, if the latter's time steps are small enough. However this system is a complex set of coupled ODEs displaying distinctly non-linear dynamics and feedback, and this could create differences between the methods, which could partially explain differences in **Figure 4 (e)** compared to their equivalent.



(a) W = 100, B = 0.01, R = 7*d **(b)** W = 100, B = 0.01, R = 10*d **(c)** W = 100, B = 0.01, R = 21*d.

(d) W = 1000, B = 0.004, R = 5*d. **(e)** W = 1000, B = 0.004, R = 10*d. **(f)** W = 1000, B = 0.004, R = 21*d.

(g) W = 1000, B = 0.004, R = 5*d. **(h)** W = 1000, B = 0.004, R = 10*d. **(i)** W = 1000, B = 0.004, R = 21*d.

Figure 4: Spatiotemporal V_{mem} patterns. Top row shows outcomes with weak field sensitivity (low W and high B), and the bottom row with strong field sensitivity (high W and low B). The columns represent different field action ranges R, from left to right, short, intermediate and long ranges. Plots are recorded for final state at $t = 5s$ with 5001 output points. Bottom row (g) – (i), is simulation with same parameters as (d) – (f) run for a shorter $t = 2s$. Shades represent cell V_{mem} states, with darker representing hyperolarised and lighter representing depolarised cells respectively. For these simulations: $v_{th} = -20mV$, $E_p = -60mV$, $E_d = -5mV$, $z = 3.0$, and $G = 0.03$

The initial conditions for **Figure 4** simulations set all cells to $-60mV$. It can be seen that the most heterogeneous spatial patterns of V_{mem} occur for strong field sensitivity (high W and low B) and intermediate field action range, R. This was expected to correlate with the highest TSE complexity as well, which was later corroborated in **Figure 5**. **Figure 4 (a) – (c)** show that increasing field action range with weak field sensitivity, results in increasing the strength of the spatial pattern, comprising hyperpolarized center and depolarised peripheral cells. Strong field sensitivity in **(d) – (f)** resulted in more complex V_{mem} patterns, while this is maximized for intermediate field action range **(e)**, especially with slight variations of initial parameters (specifically $B = 0.002$) where it is overridden for larger field action ranges **(f)** resulting in a simpler pattern resembling a more spatially defined version of **(c)**. The complexity corresponding to **(e)**, can be seen in **Figure 5** middle row, final column, with $R = 10$. This along with the general trend of strong field sensitivity increasing complexity in **Figure 5**, demonstrates that complex spatial V_{mem} patterns do translate to higher complexity measures. **Figure 4 (g) – (i)**, allow visulaisation of the system in an earlier snapshot of what becomes **(d) – (e)**, and **(h)** demonstrates qualitatively similar spatial patterns to that reported in the original paper,[3] suggesting

that the previously used fixed timestep evolution method, corresponded to less than the $5s$ simulations that had been run up until this point.
This suggests that intermediate field ranges may provide a balance between local and long-range cellular interactions, allowing distinct regions of the lattice to develop different membrane potentials while remaining electrically coupled. At larger field ranges, the increased coupling between distant regions may promote more spatially uniform patterns, reducing heterogeneity.
However, despite the correlation seen between the spatial heterogeneity in **Figure 4** and TSE complexity in **Figure 5** for given parameter values, spatial heterogeneity alone does not necessarily imply high TSE complexity, since TSE also depends on the statistical relationships between cellular states over time. The final membrane potential distributions therefore provide a qualitative visual comparison with the complexity measurements, rather than a direct measure of complexity themselves.

Full model TSE complexity analysis

Having broadly reproduced the actual membrane potential simulations of the 11×11 cell lattice, we next analysed TSE complexity again, by sweeping across the parameters G, W, B, and R. Importantly for the larger lattice, not only did this measure become more interesting in general, but we now had an extra parameter, field range R to simulate. We plotted TSE complexity against field range for each combination of W and B, with R as the independent variable. Higher TSE complexity indicates a greater balance between integration and segregation in the lattice's dynamics, whereas low complexity is associated with more independent or highly coordinated behaviour.
The TSE calculation was extended to the 121 cell lattice by using the general TSE equation for subsystems of different sizes as shown in equation **(7)**. In the 2 cell model TSE only requires three entropy calculations: one for each cell and one for the combined sytem. The lattice model calculates the average subsystem entropy for subsystems ranging in size from 1 to 60 cells, corresponding to half the total number of cells in the lattice. This is implemented the `calculate_TSE()` function, using a loop over subsystem size K. The contribution from each size is calculated by subtracting the corresponding fraction of the full system entropy from the average subsystem entropy. These contributions are then summed to obtain the approximate TSE of the lattice.
Since calculating the entropy for every possible subsystem is computationally impractical (number of subsytems containing 60 cells is $\approx 10^{35}$), an approximation method was implemented by randomly sampling 20 subsets for each subsystem size. For each subsystem size rng.choice(n, k, replace=False) randomly selects 20 subsets containing k distinct cells, and ensures that no cell is selected more than once in each subset. Twenty subsets per subsystem size were selected as a compromise between computational cost and the accuracy of the estimated average entropy. NumPy's random number generator was used with a fixed random seed to ensure reproducibility and comparability across parameter combinations. Because the numbers generated by NumPy are actually pseudo random, setting a seed, allowed comparisons between multiple sweeps of TSE complexity for different parameter values, ensuring that any differences observed were coming from the changes in parameters and not changes in the way the random numbers were generated. The above is basically a Monte Carlo approximation, where the accuracy depends on the number of samples, so time permitting more accurate simulations could be run with more smaples.
The Shannon entropy calculation (`entropy_from_states()` function) was optimized using NumPy's `packbits()` function, which compresses binary states into bytes, before identifying unique states and calculating their probabilities. This reduces the amount of data that must be processed in identifying unique states, improving efficiency.
Unlike the 2 cell model, which used each cell's median voltage as its own unique threshold for binarisation, the lattice model used a fixed threshold of $-27mV$ for all cells. This is the value of the threshold voltage (v_{th}) deifned at the start of both model's code. Each cell is therefore assigned a state of 1 when its membrane voltage is greater than or equal to $-27mV$, and 0 otherwise. The resulting binary voltage histories were transposed using NumPy's $.T$ attribute, so that each row represented the complete state of the lattice at a particular time point.
Finally, the parameter sweep was extended to include the electric-field interaction range R, alongside the field-transduction parameters W and B, producing 125 simulations , (5 per graph in **Figure 5**) these results were plotted as 25 panels, allowing the relationship between field range and TSE complexity to be investigated.

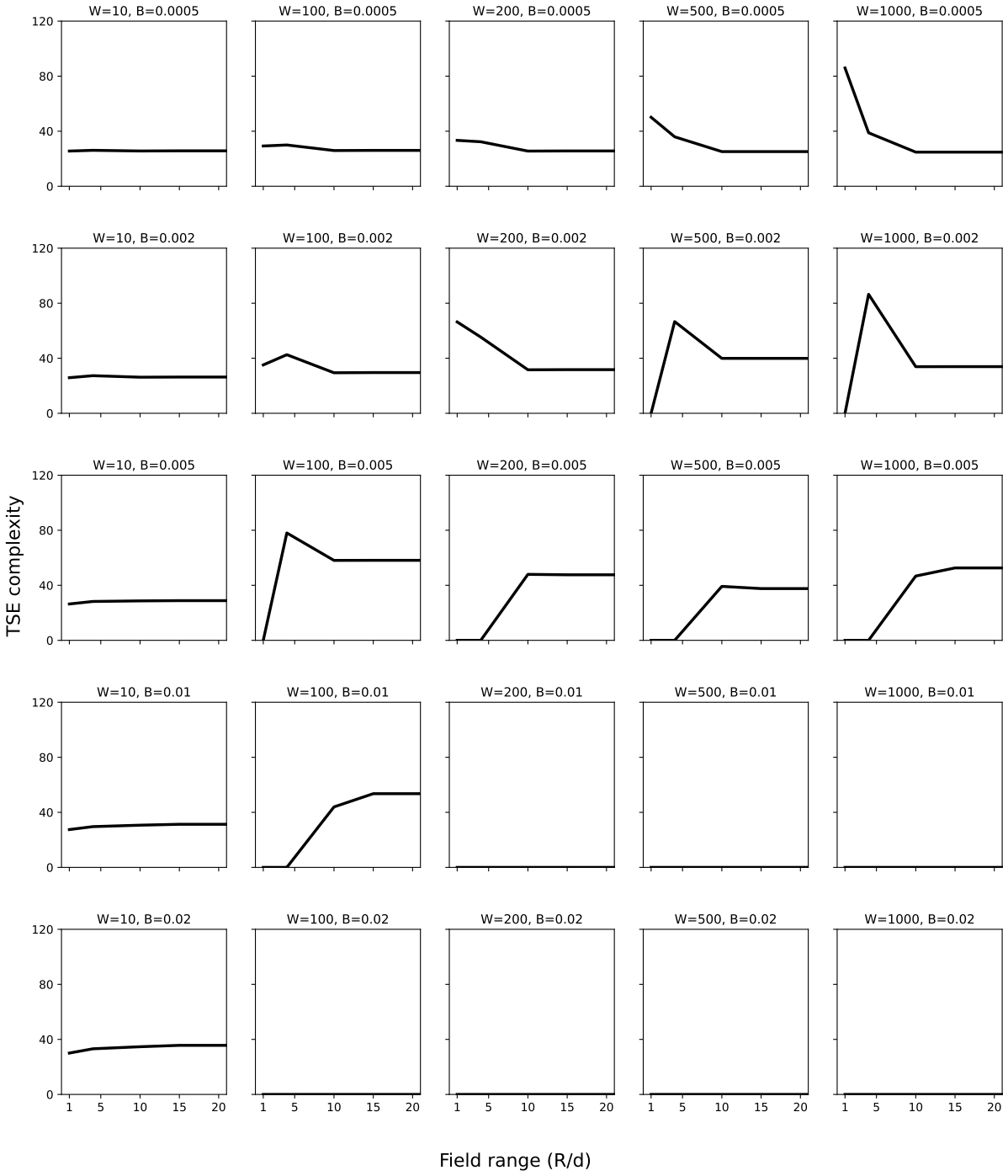


Figure 5: TSE complexity of V_{mem} patterns in 11×11 cell lattice Each plot is for different values of W and B, and TSE is plotted as a function of R. Increasing W from left to right, and increasing B from top to bottom. Weak field sensitivity bottom left, strong field sensitivity, top right. For these simulations: $v_{th} = -20mV$, $E_p = -60mV$, $E_d = -5mV$, $z = 3.0$, and $G = 0.03$

It can be seen in **Figure 5** that TSE is maximised for relatively strong field sensitivites (high W and low B), and decreases for weaker field sensitivites (low W and high B).
At weak field sensitivities, the cells are highly insensitive to the electrostatic field beyond each cell's immediate neighbors (via gap junctions), and so there is little interaction between groups of cells, and hence complexity is minimal. The flatness of the graphs for weak field sensitivity makes sense because for these parameter values, the field feedback is so weak that even when the field action range is increased significantly, it has little effect on complexity of the system. For strong field sensitivity it can be seen (top right **Figure 5**) that TSE is maximised for small values of R, and as we move down,

increasing B, this shifts to intermediate values of R and finally to high values of R in the $W = 1000$, $B = 0.005$ plot. This is because at higher thresholds (large B), a cell must be sensitive to a greater proportion of the lattice in order to be affected by its behavior. This corresponds to increasing R as we increase B (for a fixed W), in order to maximise TSE complexity. This suggests that TSE complexity is maximised when electric-field interactions provide enough coupling to coordinate cellular behavior without producing excessive synchronization. At larger field ranges, stronger collective coupling may reduce the diversity of cellular states, or at least will not increase it, and this either decreases or stabilizes complexity. The calculated TSE values are also dependent on the discretization of membrane potentials and the random sampling of cellular subsets. Therefore, differences in these methodological choices could contribute to discrepancies with the previously reported original results, where larger samples were averaged.
These results strongly agree with the previous work, except for an outlier in the $W = 100$, $B = 0.01$ panel in **Figure 5**. The most likely explanation for this is a difference in the way membrane potentials are descritized for the TSE calculation. This is because small changes in the binarisation threshold can significantly alter the measured complexity, especially near transitions between polarised and depolarised cell states.
When the field sensitivity is very weak, this approaches the behavior of the model as if there was no electrostatic field; therefore the effect in **Figure 4** and **5**, demonstrates that adding an embedded electrostatic field to the bioelectric tissue model, increases the capacity for V_{mem} patterns to form, and increases the complexity in the system. This potentially adds another layer on top of chemical organizers and gradients, which can facilitate morphogenetic information at the intercellular level, for example in embryo prepatterning and limb and organ regeneration.

Electric field mediated non–local causal interactions between cells

Long distance distance signaling is crucial in biological systems, whether in morphogenesis or in preventing disease states.[16] While many processes are well studied in facilitating long range signaling, here we study the role of electric fields in this capacity. Electric fields are known to play a role in both development and regeneration through mediating ion channels, cell migration, and proliferation.[16],[17] In this section we study whether weak electric fields can also facilitate direct cell to cell communication at relatively long distances, specifically between cells which are not directly connected. Following Manicka et al., we consider two measures, causal strength (CS) and causal distance (CD). CS measures the sensitivity of the V_{mem} of one cell to small changes in the V_{mem} of another cell, which is separated in space, and where the perturbation of the second cell occurs prior to measurement of the first. CS is calculated as the absolute value of the derivative of the V_{mem} of three representative target cells, at representative time points, with respect to the V_{mem} of every cell at $t = 0$, averaged over target cells and time, and is in dimensionless units.
CD is the distance at which CS is strongest, averaged over target cells and time, and expressed in units of distance.
The equation describing CS is:

$$CS = \frac{1}{(n_c)^2 T} \sum_{i=1}^{n_c} \sum_{j=1}^{n_c} \sum_{\tau=1}^{T} \left| \frac{\partial v_i(\tau)}{\partial v_j(0)} \right| \tag{9}$$

Where n_c is the total number of cells, T is the number of sampled time points, i is the target cell whose V_{mem} is measured, j is the source cell whose initial voltage is perturbed, and τ is the time point at which the response of cell i is measured.

The equation for CD is:

$$CD = \frac{1}{n_c} \sum_{i=1}^{n_c} \left[\frac{1}{5} \sum_{j \in Top5(i)} d(i,j) \right] \tag{10}$$

Where Top $5(i)$ are the five source cells with the highest $CS(i,j)$, and $d(i,j)$ is the Euclidean distance between the target cell i and source cell j.

Causal strength (CS) and causal distance (CD) are first introduced here as there is no analog for the

two cell model. The analysis was performed across the same parameter space as the TSE sweep, varying the field transduction parameters W and B and the field transduction range R. For each parameter combination, an initial unperturbed simulation was performed using `solve_ivp()`, with all 121 cells initially at $-60mV$ and their normalised polarising conductance's set to 1. The smulation was run for 2 seconds, and membrane potentials of target cells were sampled at 50 approximately equally spaced time points. Rather than analysing the V_{mem} response of every cell, three representative target cells were selected, at a corner, edge and interior positions of the lattice. This reduced the amount of data required for the analysis while still allowing the influence of spatial location to be taken into account. Causal strength was estimated using a finite difference approximation to the sensitivity of a target cell's membrane potential to a perturbation in another cell's initial V_{mem}. This was implemented by sequentially perturbing each of the 121 cells by $+0.1mV$ and rerunning the full lattice simulation. the original initial conditions were copied using NumPy's `.copy()` function before each perturbation, to ensure only one cell was modified per run. After each perturbed simulation, the voltage histories of the three target cells were extracted and compared with their corresponding unperturbed trajectories. The absolute value of the difference between the perturbed and unperturbed membrane potentials was divided by the perturbation amplitude and averaged over the 50 sampled time points using `np.mean()` this produced a sensitivity matrix containing the time averaged influence of each source cell on each of the three target cells.
The overall causal strength was calculated by averaging the sensitivity values across the three target cells and all other source cells. Despite being counted in the above equations' summation, self-influence was excluded by creating a copy of the sensitivity matrix using `.copy()` and replacing the entries corresponding the each target cell's influence on itself with `NaN`. NumPy's `np.nanmean()` function was then used to calculate the average while ignoring these entries, ensuring that the resulting measure was due to intercellular influence and not from the cell's response to its own initial perturbation. The overall causal strength was then calculated by averaging these sensitivities across all selected target cells, source cells (excluding self-influence), and sampled time points, giving the following finite-difference approximation of Equation **(9)**, modified to exclude self-influence:

$$CS \approx \frac{1}{m(N-1)T\Delta v} \sum_{i \in \mathcal{T}} \sum_{\substack{j=1 \\ j \neq i}}^{N} \sum_{k=1}^{T} \left| v_i^{(j)}(t_k) - v_i^{(0)}(t_k) \right| \tag{11}$$

In this equation, $N = 121$ represents the total number of cells in the lattice, m is the number of selected target cells contained in set $\mathcal{T}$, and $T = 50$ is the number of sampled time points t_k. The indices i and j denote the target and perturbed source cells respectively—with the condition $j \neq i$ excluding self-influence—while $v_i^{(j)}(t_k)$ and $v_i^{(0)}(t_k)$ represent the respective perturbed and baseline membrane potentials at time t_k following an initial voltage perturbation of $\Delta v = 0.1$ mV.

Causal distance was calculated using the same sensitivity matrix. For each target cell, the five most influential other cells were identified using NumPy's `np.argsort()` function, which sorts the sensitivity values in ascending order. The target cell itself was again excluded, this time by setting its sensitivity to $-\infty$ using `-np.inf` to ensure it could not be selected among the top five strongest influences. The Euclidean distances between the target cell and these five source cells were then calculated using `np.linalg.norm`, which was applied to the differences between their coordinates stored in `cell_positions`. These distances were then averaged for each target cell, and the results then averaged to obtain the overall causal distance. This was normalised by cell spacing d, to express the spatial extent of the causal influence in units of cell separation (12 µm).
Since each parameter combination required one unperturbed simulation and 121 individual perturbed simulations, the full sweep of 125 parameter combinations involved $15,250$ simulations. To manage the computational cost, seperate functions were defined for running the simulations `run_CS_simulation()` and calculating CS and CD `calculate_CS_CD()`. A checkpointing system was also employed using Python's `pathlib` library to organise result directories and NumPy functions to store and retrive sensitivity matrices. The matrices were saved after each completed parameter combination as a precaution against interrupts during long run times. `np.isfinite()` was used to allow the program to identify any previously completed calculations and skip them when restarted, avoiding unnecessary repetition of functions.
CS and CD were plotted against the normalised field range R/d for each combination of W and B, producing 25 panels. Matplotlib's `twinx()` function was used to create a second vertical axis within

each panel, allowing causal strength and normalised causal distance to be displayed together despite their different numerical scales. As in previous figures, consistent vertical axis limits were applied across the panels to allow direct comparison of how the strength and spatial extent of causal influence changed with the electric field action range and field transduction parameters.

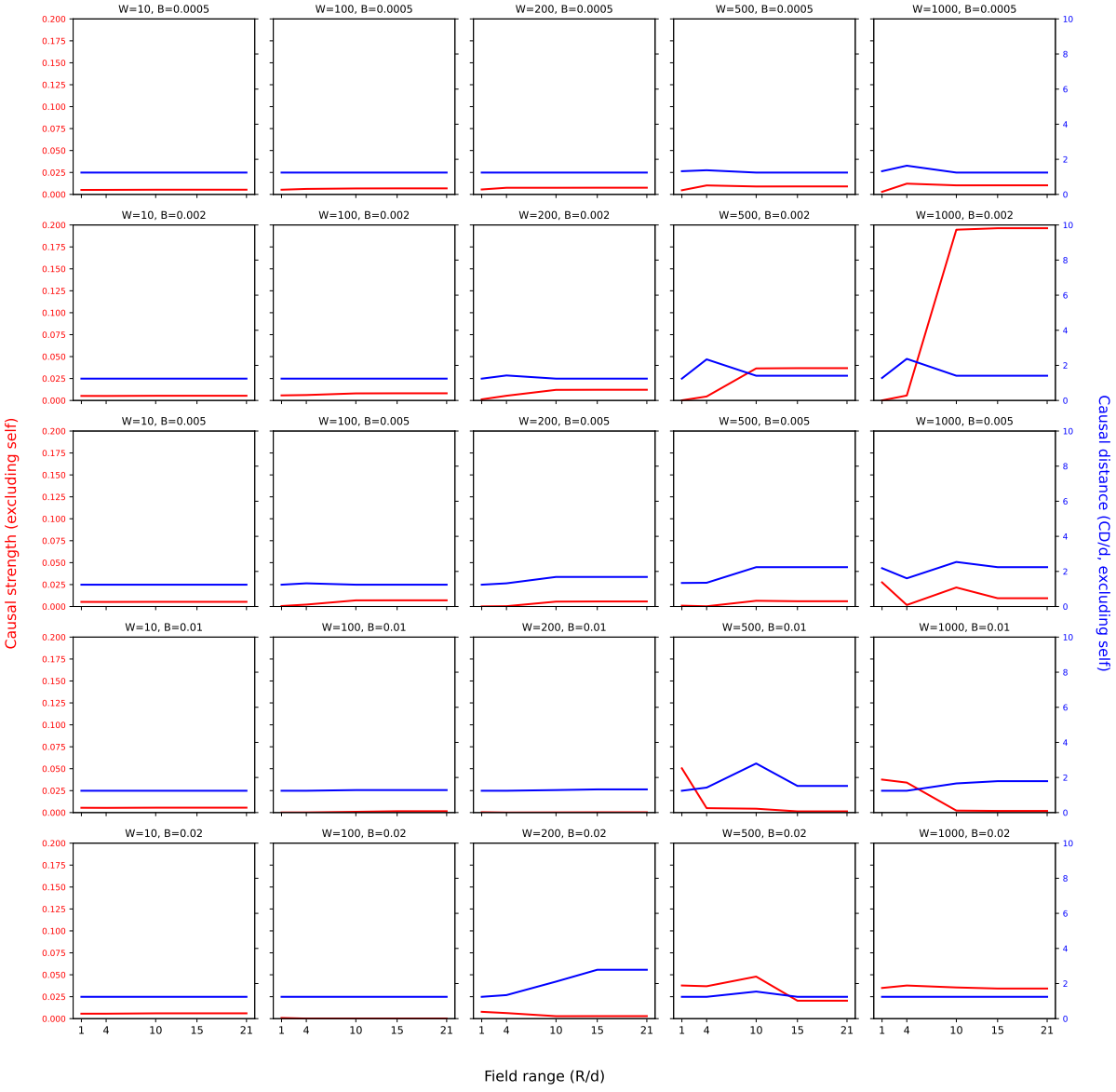


Figure 6: Causal strength (CS) and causal distance (CD): 3 target cells Red lines are (CS) plotted agains the left y-axis, blue lines are (CD) plotted against the right y-axis. For these simulations: $v_{th} = -20$ mV, $E_p = -60$ mV, $E_d = -5$ mV, $z = 3.0$, and $G = 0.03$.

These simulations show that in models with strong field sensitivity, and or large W, extremely small perturbations to the V_{mem} of a cell at the beginning of the simulation could cause changes to the V_{mem} of other cells up to 3 cells away ($\sim$36 µm), demonstrating that a cell's immediate neighbors are not necessarily the most causally influential. However, it is important to note that since CD is calculated as the average of the 5 most influential cells, there could easily be influential cells further away. The two interesting findings are that this model exhibits causal influence beyond a cell's immediate neighbors, and that for high field sensitivities and low field range the causal distance is sometimes longer than the field range. However, causal distance is far shorter than the previously reported maximum of a distance of 9 cells away.[3] This is possibly due to the fact that in this previous work, the gap junction strength was varied whereas here it was not. Despite many possible differences in computational models, this is a likely cause since gap junction coupling is a key parameter that

mediates the propagation of a perturbation through the network.
In **Figure 6**, top right and middle right panels show qualitatively similar behavior to the original work, but more generally there are large discrepancies. In particular the high CD values for strong W and B, are strikingly different, as is the behavior of CS plots in right panels of the second row, which do not reduce for larger field range R, as previously reported. At high B and W, the model can still produce strong causal interactions because W amplifies the conductance response to the electric field, even when the higher threshold B suppresses field sensitivity. .
During the initial simulations, the polarising ion-channel conductance was not constrained, allowing it to become unphysically negative. This was subsequently corrected by restricting the effective normalised polarising conductance to the interval $[0, 2]$ when calculating the membrane currents. The underlying conductance variable was allowed to evolve according to its differential equation without clipping. All figures presented here were generated using this corrected implementation.

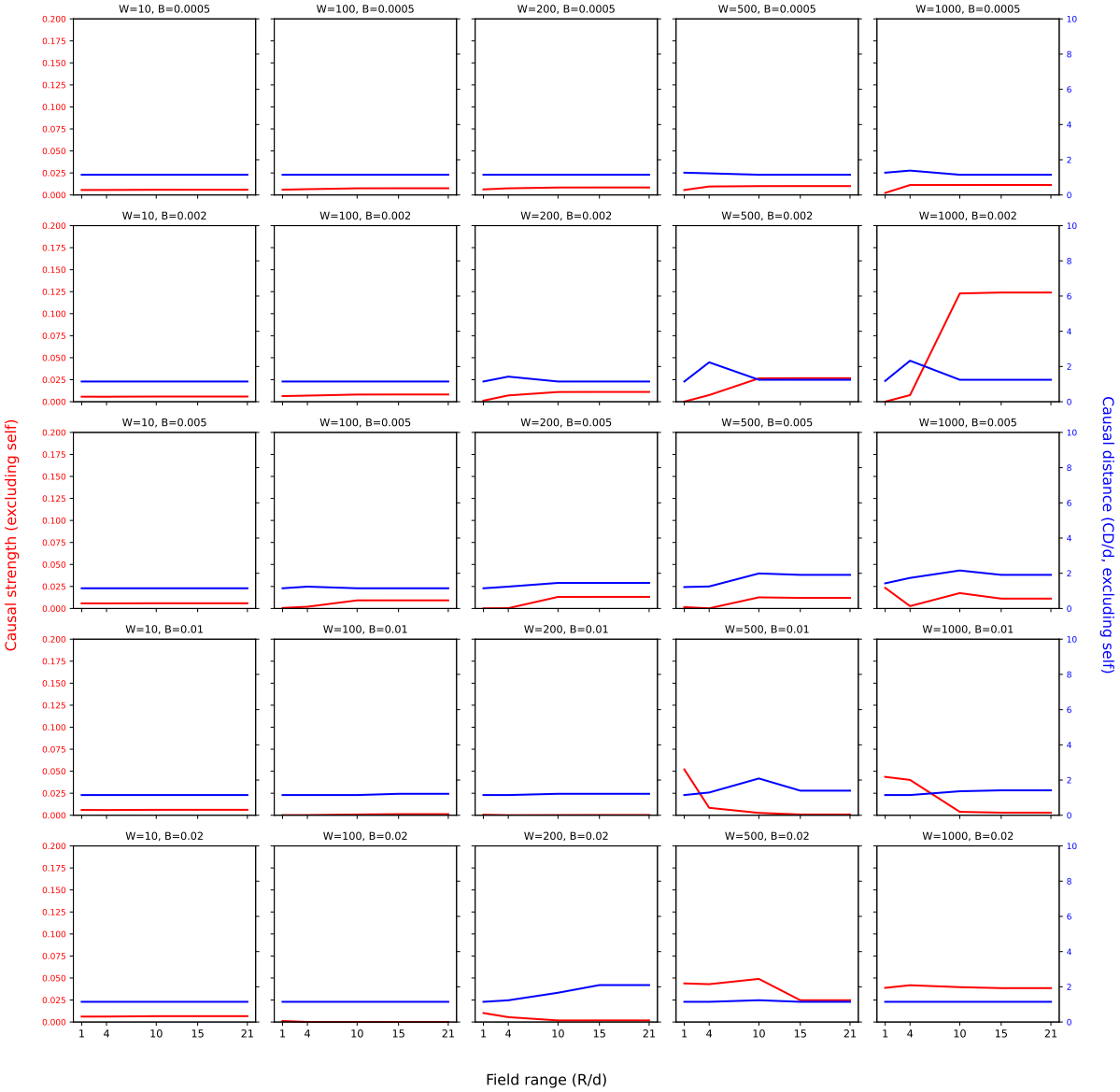


Figure 7: Causal strength (CS) and causal distance (CD): 10 **target cells** Red lines are (CS) plotted agains the left y-axis, blue lines are (CD) plotted against the right y-axis. For these simulations: $v_{th} = -20$ mV, $E_p = -60$ mV, $E_d = -5$ mV, $z = 3.0$, and $G = 0.03$.

The decision to pick three representative target cells positioned at a corner, edge and interior of the lattice, was made inline with the analysis from which this model was adapted. However this selection is not representative of the geometry of the 11×11 lattice, which has 4 corner, 36 edge, and 81 interior

cells corresponding to 3.3%, 29.8% and 66.9% of the total cells in the lattice respectively. The choice of target cells in **Figure 6**, instead gives equal 33.3% weighting to each cell position type. Because cells in different position types have different physical environements, namely the number of cells they are directly connected to via gap junctions and the amount of connectedness the surrounding field grid points have, this was hypothesized to affect their behavior and hence the CS and CD calculations.
To improve on **Figure 6**, and produce a set of more representative CS and CD values, a second sweep was performed. Utilizing the same computational model, 10 target cells were selected, with a much more representative spread of cell position types in the lattice: one corner, three edge and six interior cells, corresponding to 10%, 30% and 60% respectively. Importantly in this model, increasing the number of target cells does not increase the number of perturbed simulations, and therefore does not increase the computational time significantly.

The results in **Figure 7** show that causal strength (CS) is strongly dependent on field sensitivity, with the largest values generally occurring at higher W. In particular, for $W = 1000$ and $B = 0.002$, CS increases sharply with field range R, reaching approximately 0.12. A similar trend is observed in **Figure 6**, although CS reaches substantially higher values, exceeding 0.20 for the same parameter combination. At higher values of B, CS often decreases with increasing R, particularly for $W = 500$ and $W = 1000$.
However, causal distance (CD) remains relatively small, typically between 1 and 2 cell spacings, with occasional peaks around 3. Increasing the field range R can substantially increase CS for strong field sensitivities, without a corresponding increase in CD. This suggests that stronger field-mediated interactions do not necessarily produce more spatially distant dominant causal influences.

Overall, the 3 target and 10 target cell simulations exhibit broadly similar trends, although differences in CS magnitude are visible, particularly at high W. These differences may arise from the spatial distribution of the selected target cells. The 10 target cell simulation more closely represents the proportions of corner, edge, and interior cells within the lattice, which reduces sampling bias associated with the number of immediate neighbors each cell has. However, the effect of target-cell position on CS cannot be conclusively established without analysing each cell type separately.
It is important to note that since CD is calculated as the average of the five most causally influential cells, these results do not rule out longer distance intercellular communication, since they are likely heavily weighted by the immediate neighbors of the target cells.

These CS and CD distance results suggest that electric fields can facilitate non-local causal interactions between cells that do not depend linearly on spatial separation, and this could potentially play an important role in morphogenetic pattern formation and regulation, during development and regneration. Though this affect is weaker than previously reported.

Conclusion

This project investigated the role of bioelectric fields in multicellular pattern formation through computational modeling of membrane potential dynamics, emergent complexity, and causal interactions within cellular networks. Initially a simplified two–cell model was developed and simulated. Characteristic behavior of membrane potentials, gap junction conductances and electric fields sensed by each cell were analyzed and graphed. Two main behaviours where observed across parameter ranges where the model tended to exhibit convergence of membrane potentials and sensed electric fields, and ones for which these trajectories did not converge due to appropriately set initial conditions. Tononi Sporns Edelman (TSE) complexity analysis was then performed. This demonstrated expected biophysical behavior of the system, with the relevant influences from field transduction and cell level parameters. Notably, E_p and E_d were shown to be the preferred polarised/depolarised membrane states, and that increasing gap-junction coupling pushed neighboring cells toward similar voltages, and therefore tended to synchronize cells and suppress independent behavior, this reduced complexity over much of the W and B parameter space, and reflects recent experimental results.

The model was then expanded to a 11×11 cell lattice, allowing more interesting behavior and analysis, particularly by introducing a new parameter field action range, R. Simulations were performed for $5s$ and $2s$, producing plots of V_{mem} states of the lattice indicating polarized and depolarized cells. Interestingly, these demonstrated increased V_{mem} pattern heterogeneity for strong field sensitivity and intermediate to large field range R.
TSE complexity was again performed by sweeping through the large parameter range and demonstrated that complexity is maximised for relatively strong field sensitivities and decreases for weaker field sensitivities. The lattice V_{mem} state plots and TSE sweep, closely matched previously reported results, and together demonstrated that the addition of an embedded electrostatic field increased the system's complexity and heterogeneous pattern forming ability.
Finally measures of causal strength and causal distance were applied, again by sweeping through the parameter space. These produced similar qualitative behavior for some of the parameter space, and new behavior for others. The CS and CD results demonstrated that particulary for short field action ranges, and strong field sensitivities, especially large W, perturbations to a cell at the beginning of a simulation could cause changes in the V_{mem} of cells up to 3 cells away. This was lower than previously reported, and could be due to a multitude of factors including differences to caclulations of gap junction conductance, sampling size, and target cell lattice positions. The target cell selection was improved to more closely reflect the proportions of corner, edge, and interior cells within the lattice. The resulting CS and CD measurements were broadly consistent with the previous three-target cell simulations, although some differences were observed.
Future improvements would include implementing time dependent gap junction conductances to better capture dynamic intercellular coupling, increasing the number of sampled cell subsets in the TSE calculations to improve statistical reliability, and expanding the CS and CD analyses to include all 121 cells as targets. This would eliminate potential sampling bias due to target cell position. Further work could also investigate the sensitivity of causal strength (CS) and causal distance (CD) to the magnitude of the initial membrane potential perturbation. This would help assess the numerical stability of these measures and provide further understanding into how the strength and spatial extent of intercellular interactions depend on the magnitude of electrical disturbances. This could give a more detailed understanding of the network's underlying biophysical behaviour. Larger lattices and longer simulation times would also potentially improve characterization of bioelectric patterns, in more realistic tissue environments.

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